

5. CONCLUDE THE ANALYSIS

5.1 Model Evaluation

The Linear Regression model was evaluated using **R² Score, Mean Absolute Error (MAE), and Mean Squared Error (MSE)**.

- **R² Score:** Indicates how well the selected features explain the variation in mobile prices. A higher R² value indicates that the model explains a larger proportion of the variation in prices.
- **MAE:** Shows the average absolute difference between the actual and predicted mobile prices. A lower MAE indicates better prediction accuracy.
- **MSE:** Gives the average squared prediction error. A lower MSE indicates fewer large prediction errors.
- **Slope/Coefficient:** Shows the direction and magnitude of the relationship between the selected feature(s) and mobile price.
- **Intercept:** Represents the estimated mobile price when the selected feature values are zero.

Therefore, the model's performance should be assessed using the **actual metric values obtained from your output**. If the R² score is relatively high and MAE/MSE are relatively low for the scale of mobile prices, the model provides a reasonable linear prediction. Otherwise, additional features or a different model may be required.

5.2 Insights and Discussion

1. Insights from Correlation Analysis and Scatter Plots

The correlation analysis helps identify the **strength and direction of the linear relationship** between mobile features and mobile prices.

- A **positive correlation** means that as a feature increases, mobile price tends to increase.
- A **negative correlation** means that as a feature increases, mobile price tends to decrease.
- A correlation close to **zero** indicates a weak linear relationship.

The scatter plots help visually confirm whether the relationship between the selected feature and mobile price appears linear or whether the data points are widely scattered.

2. Contribution of Selected Features

The selected features contribute to the prediction of mobile prices according to their relationships with the target variable.

For example, if a feature has a positive regression coefficient, an increase in that feature is associated with an increase in the predicted mobile price, while a negative coefficient indicates an inverse relationship.

Features with stronger relationships with the target may provide more useful information for a simple linear regression model. However, correlation alone does not establish causation.

3. Meaning of Slope and Intercept

The regression equation can be represented as:

$$\text{Mobile Price} = \text{Intercept} + (\text{Coefficient} \times \text{Feature})$$

The **slope/coefficient** indicates how much the predicted mobile price changes when the feature increases by one unit, while keeping other features constant in a multiple-regression model.

The **intercept** is the predicted price when all predictor values are zero. Depending on the feature ranges, this may or may not have a practical real-world interpretation.

4. Predicted vs Actual Values

The difference between the actual and predicted prices represents the **prediction error**.

- Small differences indicate that the model predictions are close to the actual prices.
- Large differences indicate that the model has difficulty predicting some mobile prices.
- The **MAE and MSE** quantify these errors.
- The **R² score** indicates how much variation in mobile prices is explained by the model.

Thus, the final assessment should be based on the numerical values obtained from the model evaluation.

5. Possible Improvements

The model performance could potentially be improved by:

1. **Adding more relevant features** such as RAM, internal storage, battery capacity, camera specifications, processor type, display characteristics, and brand.
2. **Handling outliers** appropriately.
3. **Checking multicollinearity** between predictor variables.
4. **Trying multiple linear regression** if more than one feature is relevant.
5. **Testing non-linear models** if the relationship between features and price is not linear.
6. Comparing Linear Regression with models such as **Decision Tree, Random Forest, or Gradient Boosting**.
7. Performing **feature selection** to retain the most informative variables.
8. Using **cross-validation** to obtain a more reliable estimate of model performance.

Final Conclusion

The analysis demonstrates how **correlation analysis, scatter plots, Linear Regression, and evaluation metrics** can be used to study and predict mobile prices. The model provides a quantitative relationship between the selected features and mobile price. Its usefulness depends on the obtained **R², MAE, and MSE values**, as well as how closely the predicted prices match the actual prices. Further feature engineering, additional relevant variables, and comparison with other regression algorithms could be used to improve the predictive performance.